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MAJOR PROJECT REGISTRATION FORM

Date:

Project Category:	☐ Research oriented ☐ Application oriented ☐ Product oriented ☐ Other (please specify) Specify if others:	Batch ID:	CSM-C9

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Title of the Project: **AI-Powered Cyberbullying Detection**

Abstract

The rapid growth of social media platforms has transformed the way people communicate and interact online. However, this growth has also led to a significant increase in cyberbullying, where individuals are harassed, threatened, insulted, or emotionally abused through online comments and messages. Cyberbullying has become a serious social issue, particularly among teenagers and young adults, often resulting in anxiety, depression, low self-esteem, and other mental health problems. Existing moderation systems primarily rely on keyword-based filtering or traditional machine learning techniques, which often fail to detect context-dependent, sarcastic, or implicit forms of cyberbullying. As a result, many harmful comments remain undetected, highlighting the need for a more intelligent and context-aware detection system.

This project proposes an **AI-Powered Cyberbullying Detection System** using **Bidirectional Encoder Representations from Transformers (BERT)** integrated with **Explainable Artificial Intelligence (XAI)** techniques. The proposed system preprocesses social media comments, tokenizes the text using the BERT tokenizer, and classifies the comments into categories such as normal, offensive, hate speech, or cyberbullying. Unlike conventional approaches, BERT understands the contextual meaning of sentences rather than relying solely on offensive keywords, enabling the detection of indirect and context-based bullying. Furthermore, Explainable AI techniques such as SHAP or LIME are incorporated to provide transparent predictions by highlighting the words and phrases that contribute to the model's decision, thereby improving user trust and system interpretability.

The performance of the proposed model will be evaluated using standard metrics including Accuracy, Precision, Recall, F1-Score, and Confusion Matrix, and compared with traditional machine learning and deep learning algorithms to demonstrate its effectiveness. The project focuses on English text-based cyberbullying detection and aims to provide an efficient, reliable, and interpretable solution that can assist social media platforms, educational institutions, parents, and online community moderators in identifying harmful content at an early stage. By combining advanced Natural Language Processing with Explainable Artificial Intelligence, the proposed system contributes towards creating a safer digital environment while addressing the limitations of existing cyberbullying detection methods.

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References (Category specific)

1. T. H. Teng and K. D. Varathan, "Cyberbullying Detection in Social Networks: A Comparison Between Machine Learning and Transfer Learning Approaches," *IEEE Access*, vol. 11, pp. 55533–55560, 2023, doi: 10.1109/ACCESS.2023.3275130.

2. M. Al-Hashedi, L.-K. Soon, H. N. Goh, A. H. L. Lim, and S. M. Said, "Cyberbullying Detection Based on Emotion," *IEEE Access*, vol. 11, 2023, doi: 10.1109/ACCESS.2023.3280556.

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